

Embedded systems I / embedded programming

AA. Doxygen introduction

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Objectives

Be able to write comments in C language for auto generated documentation

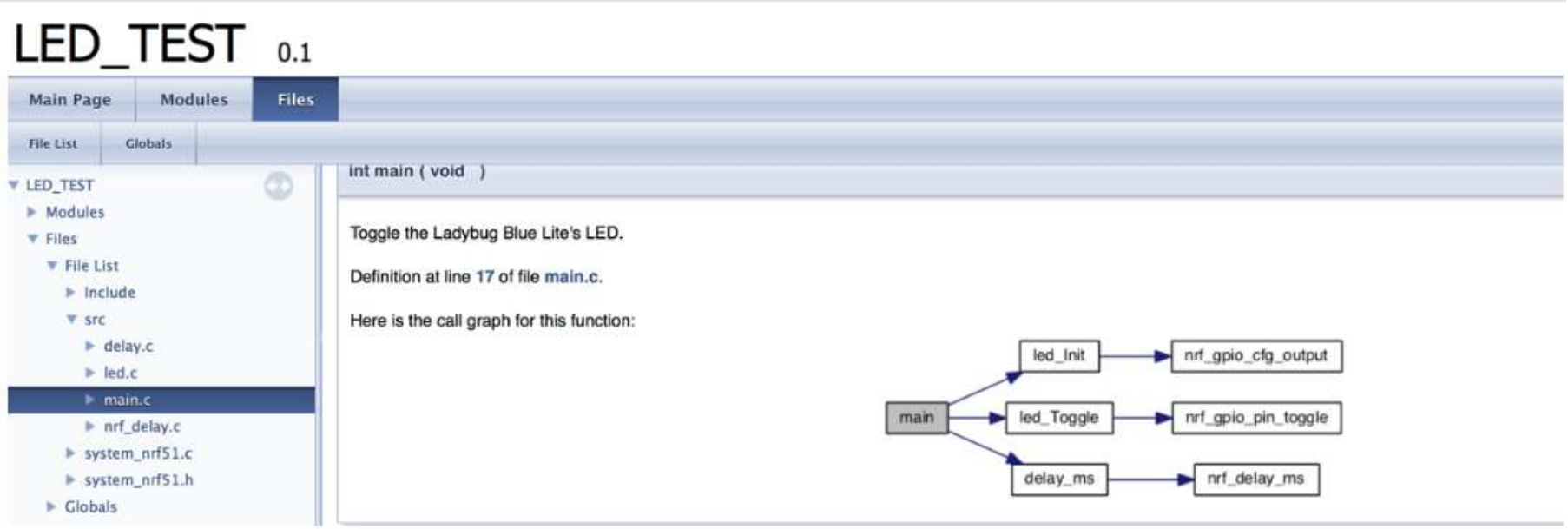
- What is Doxygen
- Features
- How to write comments
- How to build documentation
- Results

What is Doxygen

- Source code documentation generator tool
- Used for C, C++, Java, Python, VHDL, PHP, C#, Fortran, …
- Official site: https://www.doxygen.nl
- Wiki: https://en.wikipedia.org/wiki/Doxygen

Features

- Extracts documentation from source file comments
- Shows relation between the various elements (function, typedef, struct, …)
- Shows call graphs



How to write comments

- A lot of different syntax

```
/**
 * ... text ...
 */
Or
/*!
 * ... text ...
 */
Or
///
/// ... text ...
///
```

How to write comments

- Function example in a `'.h'` file

```
// ****************************************
// \brief Store graphical symbol in CGRAM LCD position
// \param graph Pointer to the graphic table
// \param mempos Memory position 0 to 7
// \return Returns false if a parameter error
// ****************************************
bool SetGraph(uint8_t * graph, uint8_t mempos);
```

How to write comments

- Variable and types documentation

```
uint8_t var; /// < Simple description

/** Enum description                                    */
 enum State {
          RUN,  ///< system is running
          IDLE, ///< system waiting event action
          STOP  ///< Needs a reboot to restart
         };
```

How to write comments

- list of keywords

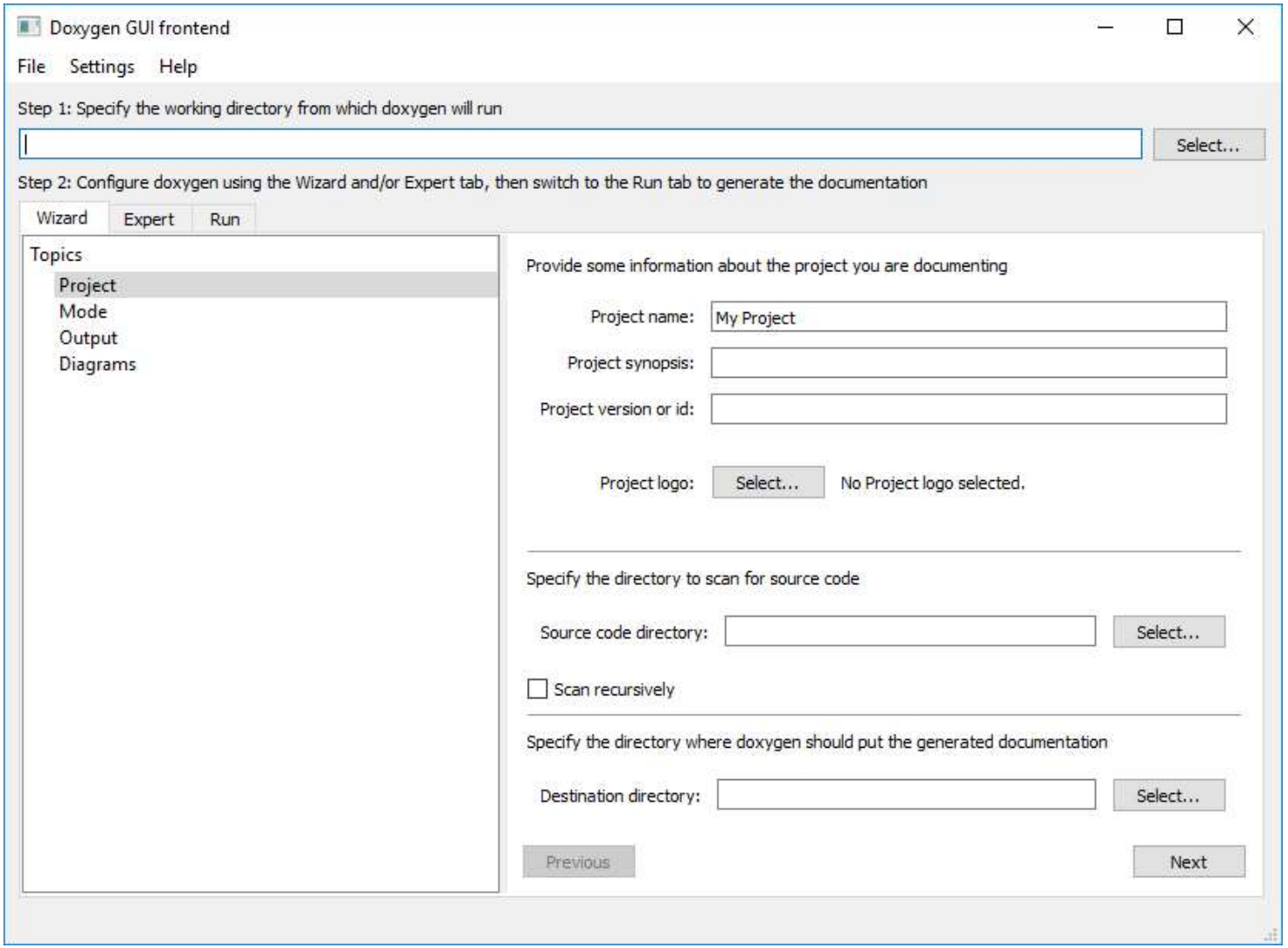
\a
\addindex
\addtogroup
\anchor
\arg
\attention
\author
\authors
\b
\brief
\bug
\c
\callergraph
\callgraph
\category
\cite
\class
\code
\collaborationgraph
\concept
\cond
\copybrief
\fileinfo
\fn
\groupgraph
\headerfile
\hidecallergraph
\hidecallgraph
\hidecollaborationgraph
\hidedirectorygraph
\hidegroupgraph
\hideincludedbygraph
\hideincludegraph
\hideinheritancegraph
\hideinlinesource
\hiderefby
\hiderefs
\hideinitializer
\htmlinclude
\htmlonly
\idlexcept
\if
\ifnot
\image
\ref
\refitem
\related
\relates
\relatedalso
\relatesalso
\remark
\remarks
\result
\return
\returns
\retval
\rtfinclude
\rtfonly
\sa
\secreflist
\section
\see
\short
\showdate
\showinitializer
\showinlinesource

\copydetails
\copydoc
\copyright
\date
\def
\defgroup
\deprecated
\details
\diafile
\dir
\directorygraph
\docbookinclude
\docbookonly
\dontinclude
\dot
\dotfile
\doxyconfig
\e
\else
\elseif
\em
\emoji
\implements
\include
\includedoc
\includedbygraph
\includegraph
\includelineno
\ingroup
\inheritancegraph
\internal
\invariant
\interface
\latexinclude
\latexonly
\li
\line
\lineinfo
\link
\mainpage
\maninclude
\manonly
\memberof
\module
\showrefby
\showrefs
\since
\skip
\skipline
\snippet
\snippetdoc
\snippetlineno
\static
\startuml
\struct
\subpage
\subsection
\subsubsection
\tableofcontents
\test
\throw
\throws
\todo
\tparam
\typedef
\union

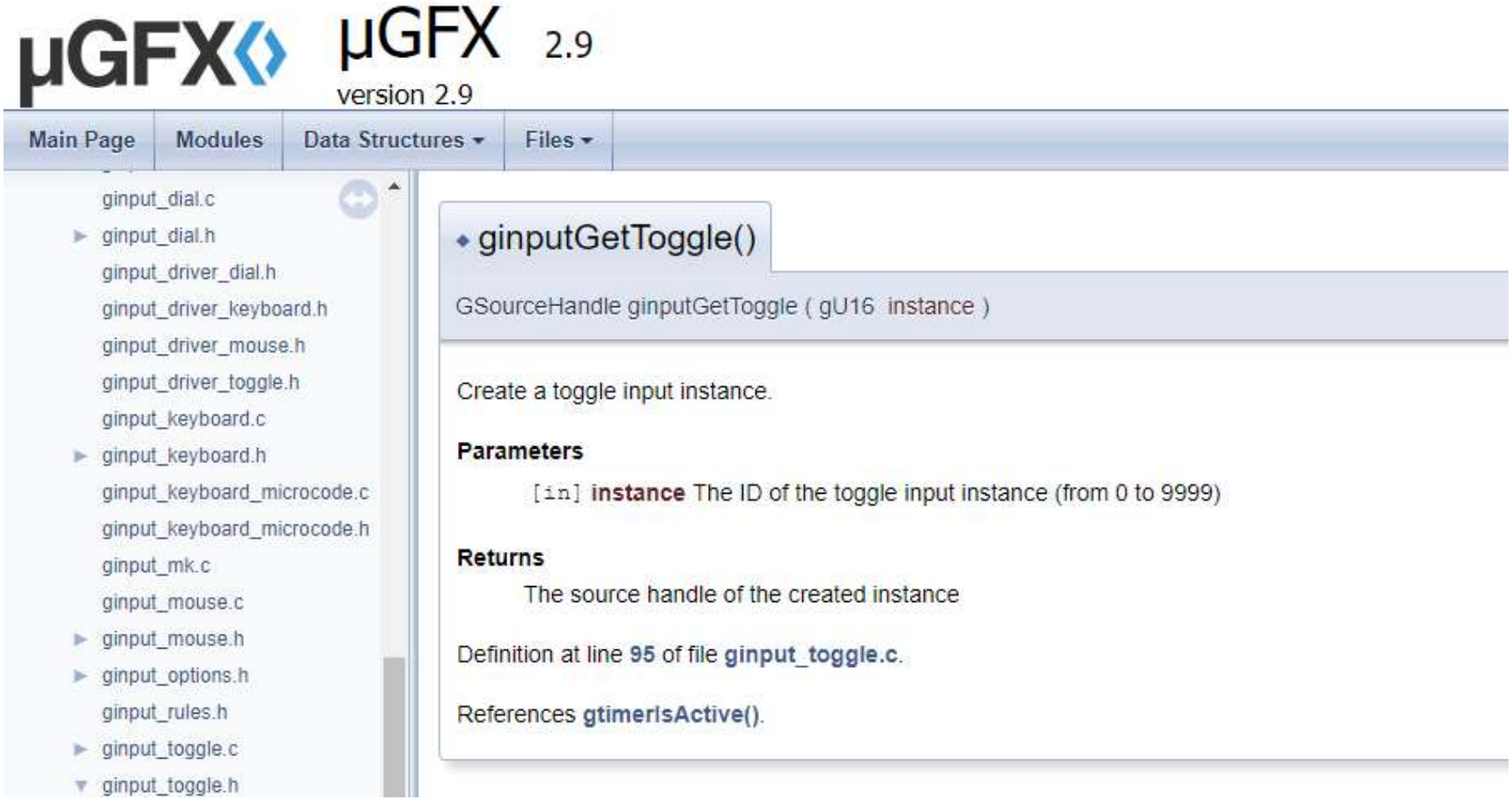
\endcode
\endcond
\enddocbookonly
\enddot
\endhtmlonly
\endif
\endinternal
\endlatexonly
\endlink
\endmanonly
\endmsc
\endparblock
\endrtfonly
\endsecreflist
\endverbatim
\enduml
\endxmlonly
\enum
\example
\exception
\extends
\f(
\f)
\f$
\f[
\f]
\f{
\f}
\file
\msc
\mscfile
\n
\name
\namespace
\noop
\nosubgrouping
\note
\overload
\p
\package
\page
\par
\paragraph
\param
\parblock
\post
\pre
\private
\privatesection
\property
\protected
\protectedsection
\protocol
\public
\publicsection
\pure
\qualifier
\raisewarning
\until
\var
\verbatim
\verbinclude
\version
\vhdlflow
\warning
\weakgroup
\xmlinclude
\xmlonly
\xrefitem
\$
\@
\\
\&
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\<
\=
\>
\#
\%
\"
\.
\::
\|
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\---

How to build documentation

- Doxy Wizard



Results


µGFX 2.9
version 2.9
Main Page
Modules
Data Structures
Files
ginput_dial.c
ginput_dial.h
ginput_driver_dial.h
ginput_driver_keyboard.h
ginput_driver_mouse.h
ginput_driver_toggle.h
ginput_keyboard.c
ginput_keyboard.h
ginput_keyboard_microcode.c
ginput_keyboard_microcode.h
ginput_mk.c
ginput_mouse.c
ginput_mouse.h
ginput_options.h
ginput_rules.h
ginput_toggle.c
ginput_toggle.h
ginputGetToggle()
GSourceHandle ginputGetToggle (gU16 instance)
Create a toggle input instance.
Parameters
[in] instance The ID of the toggle input instance (from 0 to 9999)
Returns
The source handle of the created instance
Definition at line 95 of file ginput_toggle.c.
References gtimerIsActive().